Root Cause Analysis — RCA-2025-004

NW-PowerPool-WIND-014 Substation Cascading Failures (SUB-10641 + WIN-10016 + WIN-10058)

Field	Value
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Region	NW-PowerPool
Site	NW-PowerPool-WIND-014
Assets	SUB-10641 (ABB), WIN-10016 (GE), WIN-10058 (GE)
Failure Mode Classification	E-2 (Electrical Interconnect) + S-1 (Substation Maintenance) + D-1 (Documentation Conflict)
Distribution	NW-PowerPool Operations, Reliability Engineering, Compliance, Vendor Management, CFO Office, Executive Operations Committee

Executive Summary

Three NW-PowerPool-WIND-014 assets — SUB-10641 (ABB substation interconnect), WIN-10016 (GE turbine), and WIN-10058 (GE turbine) — collectively logged **43 forced outages** and **37,557 MWh** of lost generation in FY2025 (SUB-10641: 15 outages / 13,362 MWh; WIN-10016: 16 outages / 12,589 MWh; WIN-10058: 12 outages / 11,606 MWh).

The three assets are tightly coupled: the two GE turbines feed into the ABB-supplied 34.5 kV collector and through it to the 230 kV interconnect. A substation-side fault causes turbine-side trips; a turbine-side fault stresses the substation protection scheme. Across FY2025 the coupling has been pathological.

The root cause is **a documentation conflict between ABB Substation Manual SE-7100 §6.4 (4-year maintenance cycle) and Cascadia Standard MS-2025-03 (2-year cycle) combined with GE OEM Manual GRE-2200 inspection intervals**

invalidated by the 2023 firmware refresh. Compounding factor: the Q1 2025 budget freeze deferred substation work that should have been completed in Q1 under the 2-year MS-2025-03 cycle.

1. Failure Mode Classification

E-2 (Electrical Interconnect): SUB-10641 has logged 15 forced outages including the 2025-05-02 SoC-drift event implicated in OIR-2025-006 (cascade onset) and the 2025-11-18 SoC-drift event (204 h, 2,252.8 MWh). The interconnect protection scheme has tripped on a mix of under-voltage, vibration, and SoC-drift conditions. The mechanism in each case is upstream — the protection scheme is working correctly; the underlying fault sources are the issue.

S-1 (Substation Maintenance): ABB Manual SE-7100 §6.4 specifies major maintenance every 4 years. MS-2025-03 reduced this to 2 years for substations operating in Cascadia's NW thermal cycling profile. SUB-10641 last received major maintenance in 2022-09. Under SE-7100 the next major would be due 2026-09; under MS-2025-03 it would have been due 2024-09. Cascadia operations defaulted to the OEM cycle absent a formal close-out of the conflict — i.e., MS-2025-03 §6 ("ABB Substation Manual §6.4 superseded for assets in NW-PowerPool thermal cycling profile") was not migrated into the work-order generation engine.

D-1 (Documentation Conflict): The same operational pattern applies to GE Manual GRE-2200 inspection intervals. The 2023 firmware refresh changed the load duty cycle on the GE units; the manual intervals were not updated. WIN-10058 was 1,420 hours past MS-2025-03 reinspection at the May 2025 event (OIR-2025-006). WIN-10016 was 2,640 hours past at the April 2025 event (OUT-2025-345, 2,418.4 MWh).

2. Fishbone / Contributing Factors

Method:
- ABB Manual SE-7100 §6.4 vs MS-2025-03 conflict. MS-2025-03 acknowledges the override but the WO-generation engine still queries the OEM-cycle field. CAR-2025-010 (proposed) covers the engine update.
- GE Manual GRE-2200 inspection intervals predate the 2023 firmware refresh. CAR-2025-003 has been open since 2023; this RCA recommends closing it with a unilateral Cascadia operating standard.

Machine:
- SUB-10641 transformer-bank cooling system shows progressive efficiency loss. Heat-exchanger inspection at 2024-09 last major maintenance recorded "within tolerance"; no quantitative measurement was taken.
- WIN-10016 and WIN-10058 pitch-system sub-components are 4.5 years in service;

the manufacturer-recommended replacement window is 5 years. Replacements have not been scheduled.

Manpower:
- Q1 budget freeze impacted substation surveying. The annual thermal-imaging survey (WO-2025-02077) was deferred to Q3 and ultimately completed 2025-09-04. The September survey identified a hot spot on the SUB-10641 B-phase disconnect that was action-required; the action was delayed by the Fall Overheating Sequence (RCA-2025-002).
- FMP-2 fatigue violations at NW-PowerPool-WIND-014 during the Spring Vibration Cascade (RCA-2025-001) hit this site harder than -024 or -003 because the lead techs were shared across all three sites.

Materials:
- ABB-supplied disconnect switch contacts on the B-phase show pitting consistent with arcing under repeated trip-reclose sequences. The contact set is original equipment.

Money:
- The 14 deferred Q1 PM WOs included 4 specifically scoped to SUB-10641 and the two GE turbines.

3. Timeline of Relevant CARs and Outages

Date	Event	Reference
2023-04	GE firmware refresh deployed at WIND-014	(GE bulletin)
2023-05	CAR-2025-003 opened (under prior numbering, re-numbered Feb 2025)	CAR-2025-003
2025-02	Standard MS-2025-03 issued; supersedes ABB and GE OEM cycles	MS-2025-03
2025-04-01	WIN-10016 overheating event (175 h, 1,932.5 MWh)	OUT-2025-336
2025-04-27	WIN-10016 soc_drift (219 h, 2,418.4 MWh)	OUT-2025-345
2025-05-02	WIN-10058 forced outage opens Spring Vibration Cascade	OIR-2025-006
2025-06-28	SUB-10641 efficiency_decline (176 h, 1,943.6 MWh)	OUT-2025-1627
2025-09-04	Annual thermal-imaging survey identifies SUB-10641 B-phase hot spot	(WO-2025-02077 close-out)

Date	Event	Reference
2025-11-18	SUB-10641 soc_drift (204 h, 2,252.8 MWh)	OUT-2025-1632
2026-02-18	This RCA issued	RCA-2025-004

4. Quantified Business Impact

Item	Value (USD)
FY25 Apex invoicing across the 3 assets (43 events)	~$3,200,000
FY25 CPS invoicing	~$640,000
Total FY25 vendor cost	**~$3,840,000**
Lost generation revenue (37,557 MWh × $60)	$2,253,420
FY25 total financial impact at WIND-014	~$6.09M

For context, the NW-PowerPool stated FY25 net profit is $103.5M and stated maintenance cost is $13.4M. The three assets at this single site account for ~29% of NW-PowerPool's stated annual maintenance spend, with the substantial majority of that spend in emergency invoicing rather than preventive work.

5. Long-Term Corrective Recommendations

1. **Update the WO-generation engine** to query MS-2025-03 override fields rather than the OEM cycle fields. Track under CAR-2025-010 (proposed). Without this fix, the documentation conflict will continue to drive deferral patterns.
2. **Close CAR-2025-003** with a unilateral Cascadia operating standard for GE units operating with the 2023 firmware refresh. Open a quarterly engagement with GE Renewable Energy to push for an updated manual; do not wait on GE.
3. **Execute SUB-10641 major maintenance** within Q1 2026. This is now 18 months overdue under MS-2025-03 and should not slip further. RAM-2025-008 in preparation.
4. **Schedule pitch-system overhaul on WIN-10016 and WIN-10058** within Q2 2026. Manufacturer-recommended 5-year window will be exceeded mid-2026.
5. **Add a second NW-PowerPool service depot** (Pendleton or The Dalles). The single-depot constraint at Bend continues to drive EPA-2024-11 activations of Apex.
6. **Cross-reference RCA-2025-001 (Spring Vibration Cascade) and RCA-2025-002 (Fall Overheating Sequence).** The WIND-014 site appears in both

prior RCAs as a contributing site; the failure pattern is now site-specific enough to warrant standalone treatment.

6. Vendor Performance Commentary

ABB Power Grids (Hitachi ABB Power Grids) Manual SE-7100 is a high-quality document for typical service profiles. The 4-year cycle is appropriate for grid-side substations in steady thermal environments. The mismatch with NW-PowerPool's thermal cycling profile is a documentation gap — Cascadia should engage ABB to issue a service bulletin acknowledging the NW thermal-cycling override.

GE Renewable Energy OEM Manual GRE-2200 inspection intervals continue to lag the 2023 firmware refresh. This has been an open issue for 2+ years. Cascadia has been carrying the cost of the gap.

Apex Emergency Grid Solutions invoiced approximately $3.2M against the three assets in FY2025. Repeat-vendor pattern reinforces RCA-2025-001 cost-arbitrage findings.

Cascadia Power Services Inc. has the engineering capability to execute the SUB-10641 major maintenance; the constraint is depot footprint and scheduling, not technical competence.

Signatures

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References: OIR-2025-006 · CAR-2024-118 · CAR-2025-002 · CAR-2025-003 · CAR-2025-010 (proposed) · RAM-2025-008 (draft) · EMR-2025-003 · RCA-2025-001 · RCA-2025-002 · RCA-2025-003 · ABB Substation Manual SE-7100 · GE OEM Manual GRE-2200 · Standard MS-2025-03 · CSM-7 · FMP-2 · EPA-2024-11